Cloud-Based Data Analysis Project

(Azure + Google Cloud Platform + Looker Studio)

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Role: Data Analyst Intern

Tools Used: Azure, Google BigQuery (GCP), Looker Studio

Project Architecture Overview

Data Flow Pipeline:

Data Source → Azure Storage → BigQuery → SQL Analysis → Looker Studio

Explanation:

Data was stored in Azure Cloud storage
Processed and analyzed using BigQuery (GCP)
SQL queries were used to generate insights
Results were visualized using Looker Studio

Key Concepts:

This project demonstrates an end-to-end cloud data pipeline involving:

- Data Storage
- Data Processing
- Data Visualization

Azure Cloud Implementation

Objective:
To understand and implement cloud-based data storage using Microsoft Azure

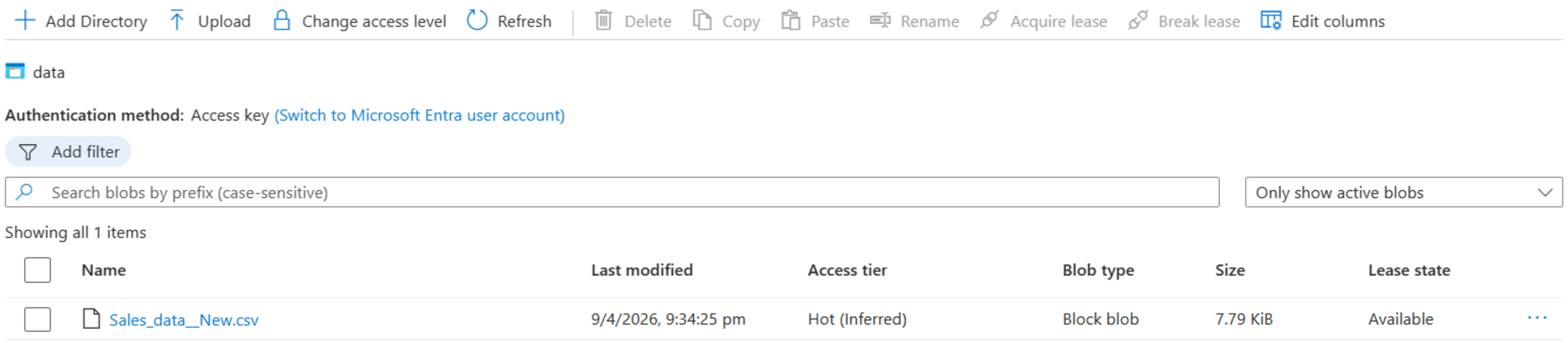
Tasks Performed:

- Created a cloud storage container in Azure
- Uploaded dataset (CSV/Excel) into Azure storage
- Managed data in a cloud environment instead of a local system

Key Learnings:

- Gained understanding of cloud storage concepts
- Learned how to manage and upload data in Azure
- Understood how cloud platforms handle data storage

Outcome:
Successfully stored and managed dataset in Azure cloud, forming the first step of the data pipeline



Azure Storage Container Showing Uploaded Dataset

Google Cloud Platform (BigQuery Implementation)

Objective:
To perform cloud-based data processing and analysis using BigQuery

Tasks Performed:

- Created dataset and table in BigQuery
- Imported data from CSV into BigQuery table
- Executed SQL queries to analyze data

SQL Analysis Performed:

- Total sales calculation
- Sales distribution by region
- Product category performance analysis
- Customer-level sales insights

Key Learnings:

- Hands-on experience with a cloud data warehouse (BigQuery)
- Improved SQL skills for real-world data analysis
- Learned how to handle large datasets efficiently

Outcome:

Successfully analyzed sales data using SQL in BigQuery to extract meaningful business insights

Untitled query

Run | Save | Download | Share

```sql
SELECT Region, SUM(Sales_Amount) AS total_sales
FROM `internship-493614.intern_project.data_table`
GROUP BY Region
ORDER BY total_sales DESC;
```

Query completed

Using on-demand processing quota

Query results

Chat | Save results | Open in

Job information | Results | Visualization | JSON | Execution details | Exec

Row	Region	total_sales
1	South	1308100
2	West	866900
3	East	220500
4	North	167100

Reference

data_table | Preview

internship-493614.intern_project.data_table

100 rows • 10.4 KB • Last modified Apr 17, 2026, 7:45 PM

Filter | Enter property name or value

Field name	Type
Order_ID	INTEGER
Customer_Name	STRING
Gender	STRING
Age	INTEGER
City	STRING
Region	STRING
Product_Category	STRING
Product_Name	STRING
Sales_Amount	INTEGER
Quantity	INTEGER

SQL-Based Sales Analysis (Region-wise) in BigQuery

Data Visualization using Looker Studio

Objective:
To create an interactive dashboard for visualizing sales data and insights

Dashboard Features:

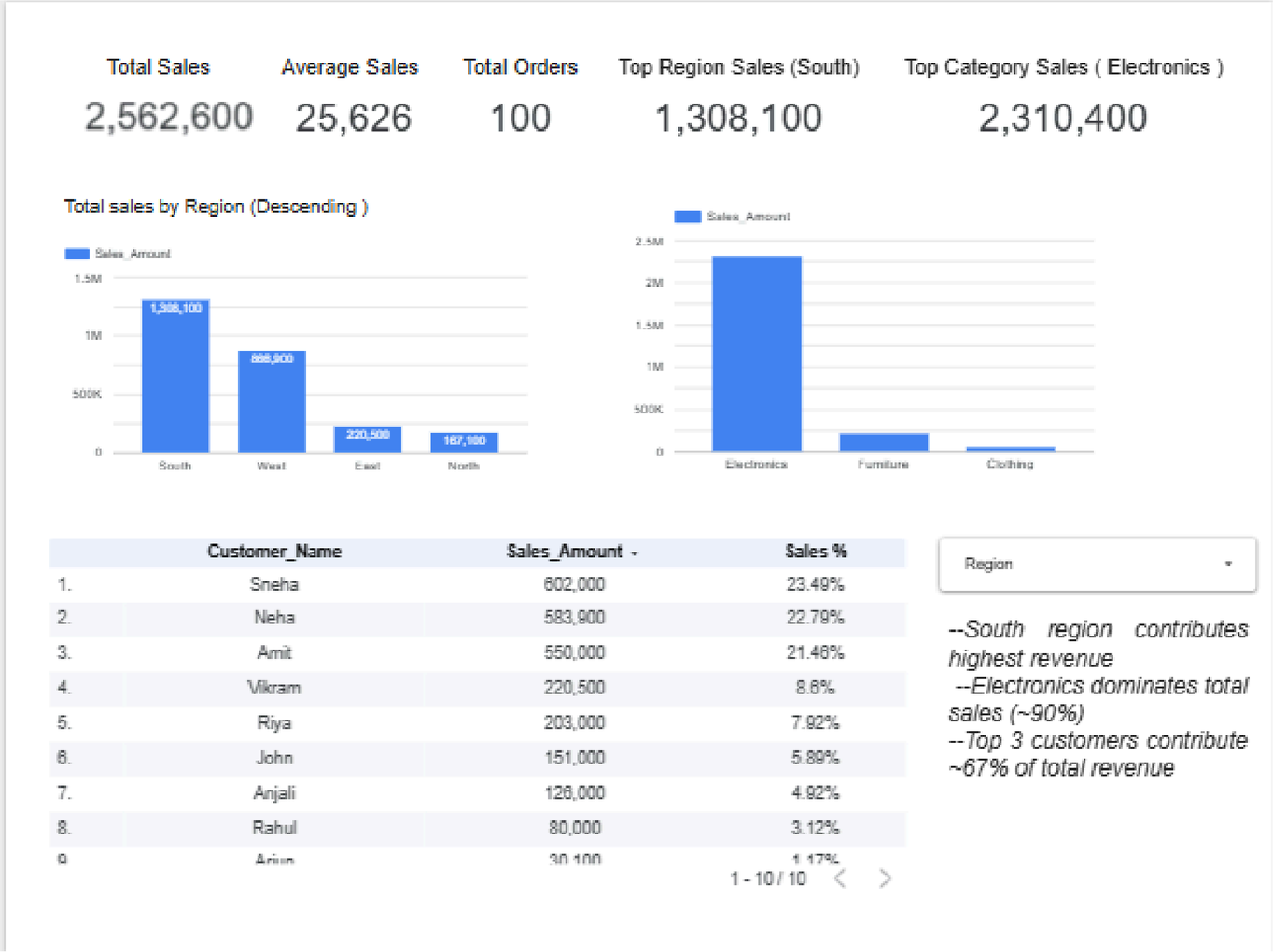
- KPI Metrics: Total Sales, Average Sales, Total Orders
- Regional Sales Analysis (Bar Chart)
- Product Category Distribution
- Customer-Level Sales with Percentage Contribution
- Interactive Filter (Region-wise analysis)

Key Insights:

- South region contributes the highest revenue
- Electronics category dominates overall sales (~90%)
- Top customers contribute a significant portion of total revenue

Outcome:

Developed an interactive dashboard to transform raw data into actionable business insights



	Customer_Name	Sales_Amount	Sales %
1.	Sneha	602,000	23.49%
2.	Neha	583,900	22.79%
3.	Amit	550,000	21.46%
4.	Vikram	220,500	8.6%
5.	Riya	203,000	7.92%
6.	John	151,000	5.89%
7.	Anjali	126,000	4.92%
8.	Rahul	80,000	3.12%
9.	Arjun	30,100	1.17%

Conclusion

Project Summary:

- Built an end-to-end cloud-based data analysis pipeline
- Used Azure for data storage, BigQuery for data processing, and Looker Studio for visualization
- Performed SQL-based analysis to extract meaningful business insights

Key Skills Gained:

- Cloud Computing (Azure & Google Cloud Platform)
- SQL for Data Analysis (BigQuery)
- Data Visualization & Dashboarding (Looker Studio)
- Analytical Thinking & Insight Generation

Key Takeaway:

This project demonstrates the ability to work with cloud-based tools to transform raw data into actionable insights, following a complete data analysis workflow

Thank you